

Loong Kuan Lee

✉ lee@lklee.dev

🐙 github.com/lklee9

🔗 gitlab.com/lklee

🆔 orcid.org/0000-0002-9967-1319

Education

2019–2023 **Doctor of Philosophy (PhD)**, *Monash University*, Melbourne

2015–2018 **Bachelor of Informatics and Computation Advanced (Honours)**, *Monash University*, Melbourne, Final Mark - 88/100

Specialised in Computer Science and Statistics & Probability. Graduated with First Class Honours.

Doctoral Thesis

title *Computing Divergences between High Dimensional Graphical Models*

supervisors Geoff Webb, Daniel Schmidt, Nico Piatkowski

year 2023

description We develop a method for computing the joint, marginal, and conditional $\alpha\beta$ - divergences, a family of divergences that include the Kullback-Leibler divergence and the Hellinger distance. We then apply this method to modifying the parameters of a decomposable model such that the resulting model is some target amount of divergence away from the original.

Honours thesis

title *Generating Concept Drift by Shuffling Instances*

supervisors Geoff Webb

year 2018

description We propose a method for shuffling the instances in a dataset such that the divergence between the empirical distribution of the first and second half of the resulting dataset reaches some target amount of divergence.

Experience

Academia

2016–2017 **Undergraduate Research Assistant**, *Monash University*, Melbourne

Researched **Concept Drift**, specifically how to measure and visualise concept drift in both streaming and static data.

Tasks:

- Developed a system to incrementally measure changes to the probability distributions of a data set over time, using **Java** and the library **Weka**.
- Developed a companion web application for the system above using **Scala** with the **Play Framework**, **Javascript**, and **HTML**.
- Used **R** extensively to visualise results and produce reports.

Industry

- 2016 **Winter Research Project**, *Agilent & Monash University*, Melbourne
Developed application to compare and analyse large groups of timeseries data over the same domain, using **R** and **Shiny** .
- 2017 **Software Tester**, *Carsales*, Melbourne
Tested backend APIs being developed to move product from a monolith to a microservice architecture. Helped develop a prototype model to predict car depreciation.
Tasks:
- Created automated tests for APIs in CI pipeline using **Postman** , **Node.js** , and **Jenkins** .
- Carried out performance testing of APIs using **Scala** and **Gatling** .
- Managed communication across multiple teams to track bugs reported by consumers of the backend APIs and bugs I found in systems the APIs depend on.
- Developed prototype to predict the depreciation of a car using **R** , **Azure ML Studio** , and **Vue.js** .

Scholarship & Awards

- 2015 Faculty of IT International Merit Scholarship
2015 Dean's Achievement Award
2016 Summer Research Scholarship — Faculty of IT
2016 Winter Research Scholarship — Faculty of IT
2017 Information Technology IBL (Industry Based Learning) Placement Scholarship
2017 Dean's Achievement Award
2019 Australian Government RTP (Research Training Program) Scholarship

Publications

Published

Loong Kuan Lee, Geoffrey I. Webb, Daniel F. Schmidt, and Nico Piatkowski. Computing Marginal and Conditional Divergences between Decomposable Models with Applications. In *2023 IEEE International Conference on Data Mining (ICDM)*, pages 239–248, December 2023 .

Loong Kuan Lee, Nico Piatkowski, François Petitjean, and Geoffrey I. Webb. Computing divergences between discrete decomposable models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(10):12243–12251, June 2023 .

Geoffrey I. Webb, Loong Kuan Lee, Bart Goethals, and François Petitjean. Analyzing concept drift and shift from sample data. *Data Mining and Knowledge Discovery*, 32(5):1179–1199, September 2018 .